Flutter Fundamentals –
User interface – Interactivity

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1. User interface

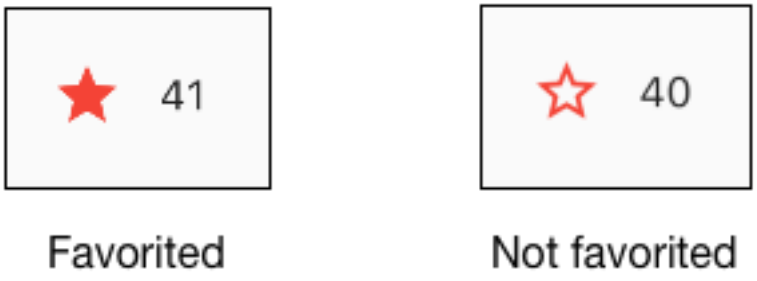
Interactivity: https://docs.flutter.dev/ui/interactivity

Key points:

1. Phân biệt: Stateful and stateless widgets
2. Các bước tạo Stateful widgets
3. Quản lý trạng thái (Managing state):
    - The widget manages its own state
    - The parent manages the widget's state
    - A mix-and-match approach
4. Other interactive widgets

Bài tập 1. Sử dụng Stateful widget xây dựng Favorite button

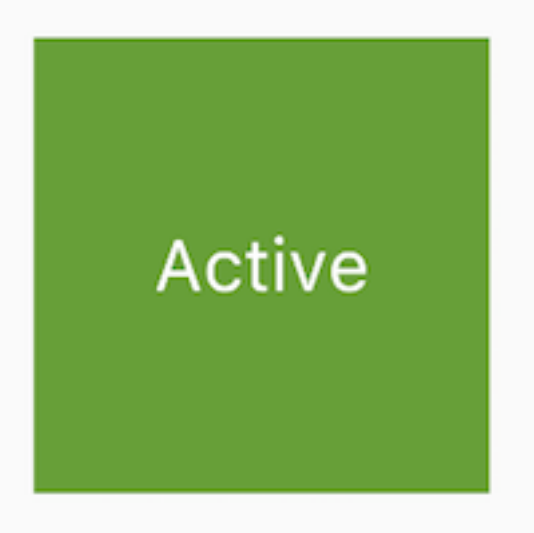
Thực hiện chức năng:



Bài tập 2. Quản lý trạng thái

Thực hiện tạo nút với 2 trạng thái: Active và Inactive theo 3 cách:

- The widget manages its own state
- The parent manages the widget's state
- A mix-and-match approach





Stateful and stateless widgets

A widget is either stateful or stateless. If a widget can change—when a user interacts with it, for example—it's stateful.

Stateless widgets

Never changes

A stateless widget never changes. Icon, `IconButton`, and `Text` are examples of stateless widgets. Stateless widgets subclass `StatelessWidget`.

Examples of stateless widgets

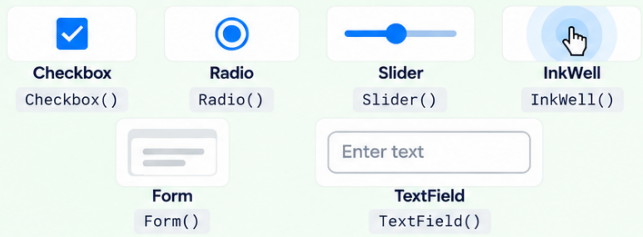


Stateful widgets

Can change (dynamic)

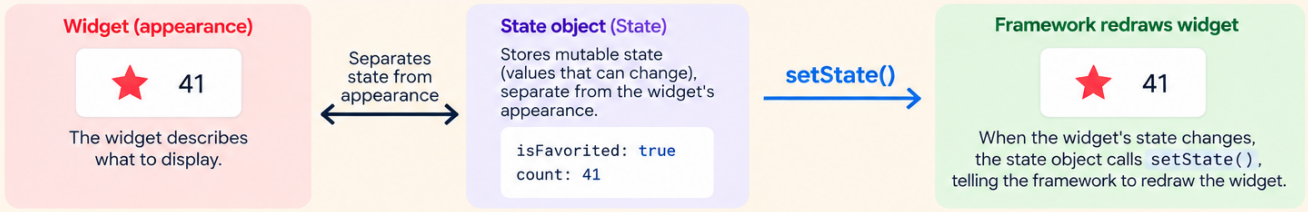
A stateful widget is dynamic: for example, it can change its appearance in response to events triggered by user interactions or when it receives data. `Checkbox`, `Radio`, `Slider`, `InkWell`, `Form`, and `TextField` are examples of stateful widgets. Stateful widgets subclass `StatefulWidget`.

Examples of stateful widgets



How state works

A widget's state is stored in a `State` object, separating the widget's state from its appearance. The state consists of values that can change, like a slider's current value or whether a checkbox is checked. When the widget's state changes, the state object calls `setState()`, telling the framework to redraw the widget.



Creating a stateful widget

What's the point?

1. A stateful widget is implemented by two classes: a subclass of `StatefulWidget` and a subclass of `State`.
2. The state class contains the widget's mutable state and the widget's `build()` method.
3. When the widget's state changes, the state object calls `setState()`, telling the framework to redraw the widget.

In this section, you'll create a custom stateful widget. You'll replace two stateless widgets—the solid red star and the numeric count next to it—with a single custom stateful widget that manages a row with two children widgets: an `IconButton` and `Text`.

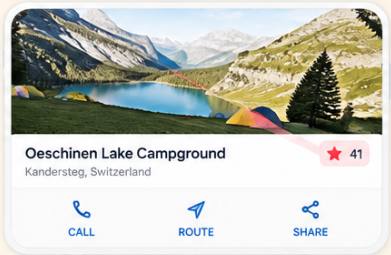
Implementing a custom stateful widget requires creating two classes:

1. A subclass of `StatefulWidget` that defines the widget.
2. A subclass of `State` that contains the state for that widget and defines the widget's `build()` method.

Example: FavoriteWidget

A custom stateful widget that manages an `IconButton` (star) and a `Text` (count).





1 Subclass of StatefulWidget

Defines the widget and creates the `State` object.

```dart
class FavoriteWidget extends StatefulWidget {
  const FavoriteWidget({super.key});

  @override
  State<FavoriteWidget> createState() => _FavoriteWidgetState();
}
```

2 Subclass of State

Contains the state for the widget and defines the `build()` method.

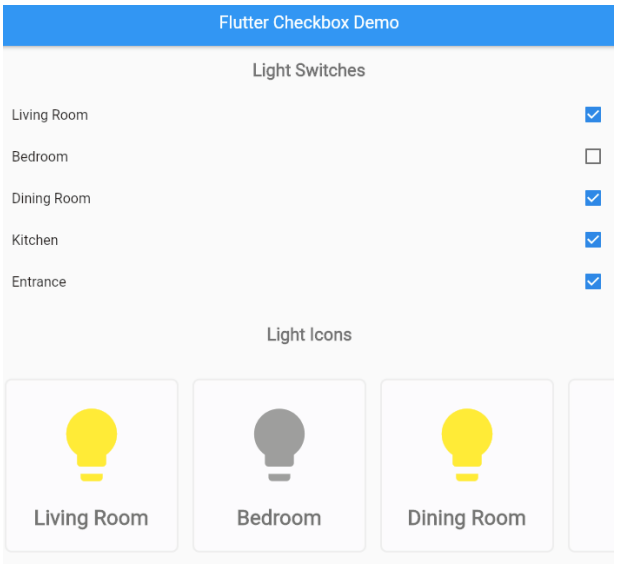
```dart
class _FavoriteWidgetState extends State<FavoriteWidget> {
  bool _isFavorited = true;
  int _count = 41;

  @override
  Widget build(BuildContext context) {
    // Return a Row with IconButton and Text
  }

}
```

2. Thực hành

Bài tập 3. Bật tắt bóng đèn với Checkbox



Bài tập 4. Thay đổi độ sáng bóng đèn với Slider

Mỗi bóng đèn sẽ có một thanh trượt Slider để thay đổi độ sáng.